



VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY  
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

# **Chapter 1: INTRODUCTION TO ALGORITHMS AND FLOWCHARTS**

September 2025

# Content

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- 
- 1 Introduction to Algorithms
  - 2 Using Flowcharts to Describe Algorithms
  - 3 Writing Algorithms in Pseudocode
  - 4 Branching (Selection) in Algorithms
  - 5 Loops (Iteration) in Algorithms
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# Content

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- |   |                                         |
|---|-----------------------------------------|
| 1 | Introduction to Algorithms              |
| 2 | Using Flowcharts to Describe Algorithms |
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| 4 | Branching (Selection) in Algorithms     |
| 5 | Loops (Iteration) in Algorithms         |
-

# Introduction to Algorithms

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- An **algorithm** is defined as a step-by-step sequence of instructions that describes how the data are to be processed to produce the desired outputs.
- In essence, an algorithm answers the question: *“What method will you use to solve the problem?”*.
- You can describe an algorithm by using flowchart symbols. By that way, you obtain a flowchart.
- **Flow chart** is an outline of the basic structure or logic of the program.
- Another way to describe an algorithm is using **pseudocode**.

# Content

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- 
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-

# Flowchart symbols

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Terminal



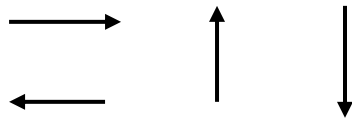
Input/output



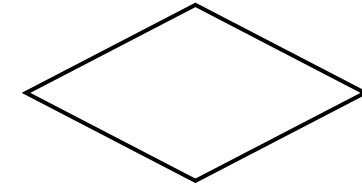
Process



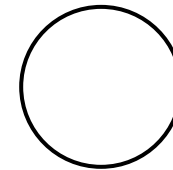
Flowlines



Decision



Connector

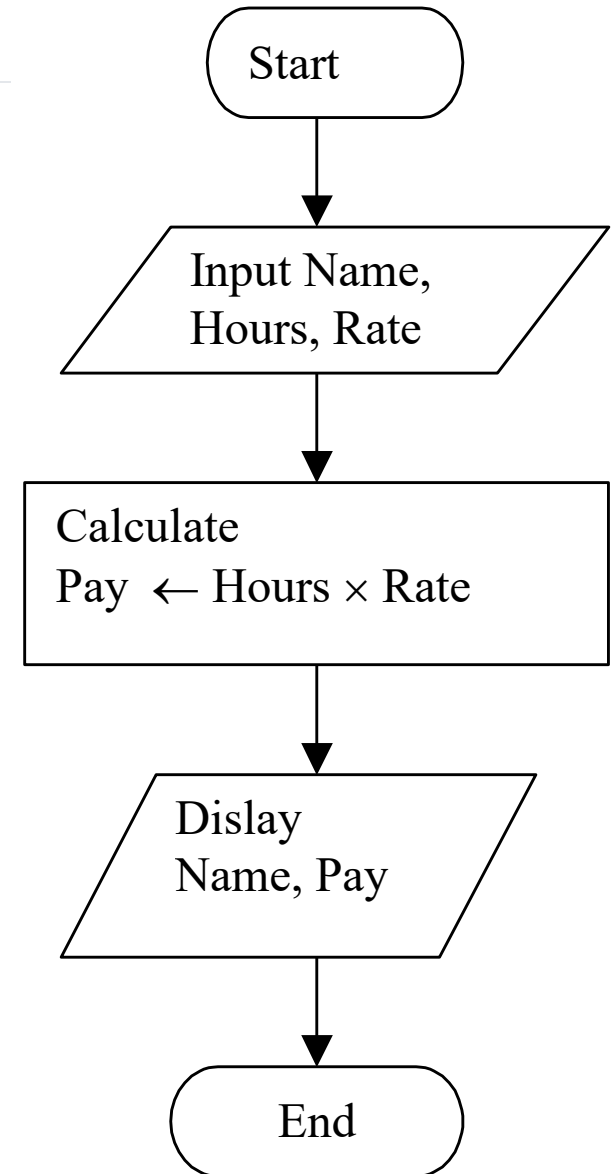


Predefined process



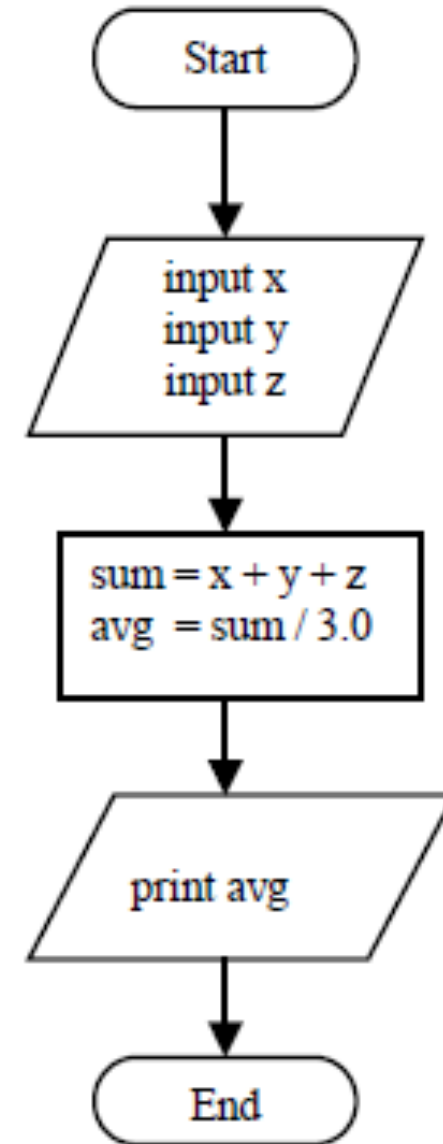
## Example 1.2.1

- Calculate the payment of an employee
- Note: *Name, Hours and Pay* are **variables** in the program.



## Example 1.2.2

- Calculate the average of 3 numbers





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- 
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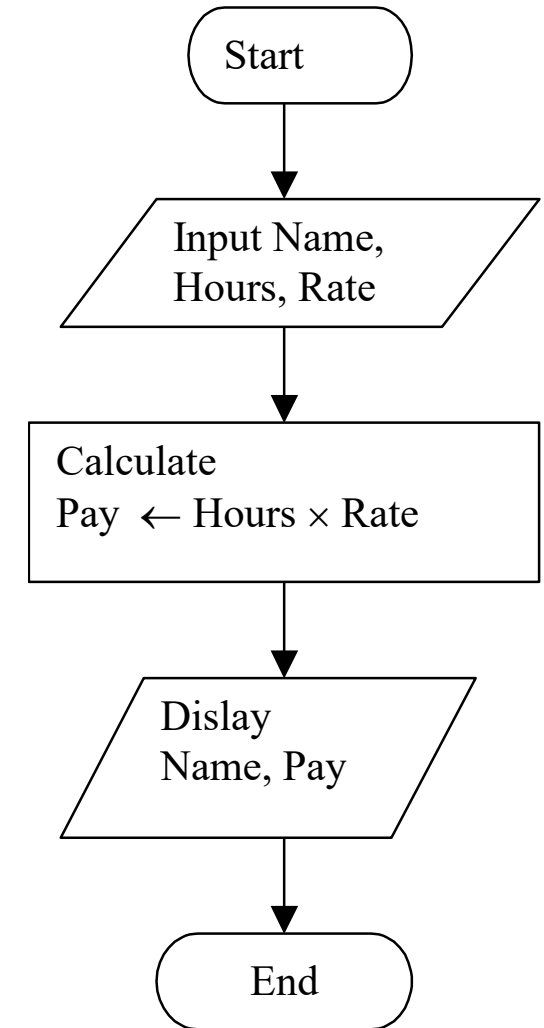
# Writing Algorithms in Pseudocode

- You also can use English-like phrases to describe an algorithm. In this case, the description is called **pseudocode**.
- The pseudocode for the program with the flowchart:

**Input** the three values into the variables Name, Hours, Rate

**Calculate**  $\text{Pay} = \text{Hours} * \text{Rate}$

**Display** Name and Pay

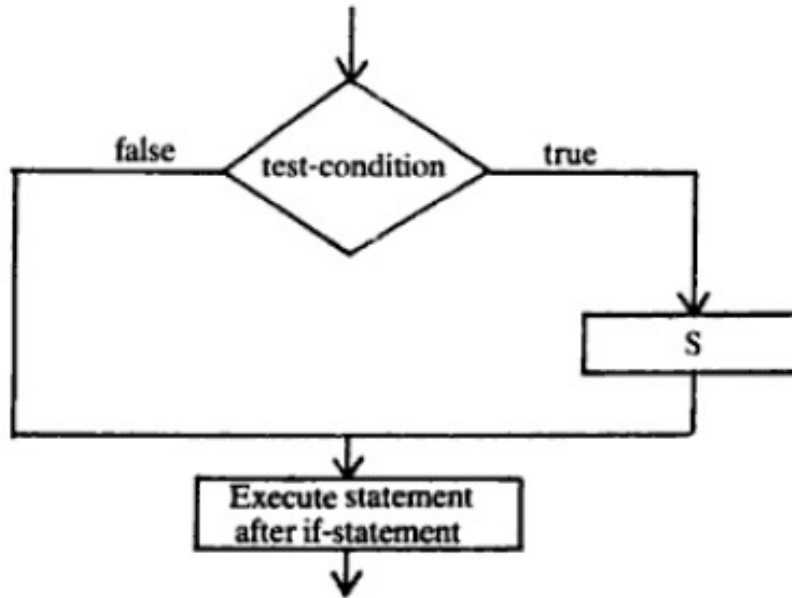


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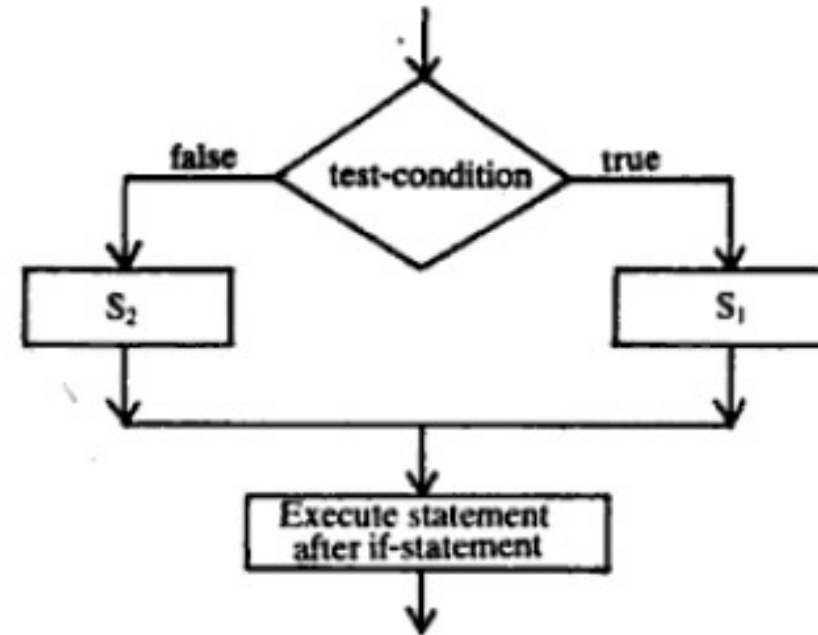
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- 
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-

# Branching (Selection) in Algorithms



**If** test-condition **then** S  
**endif**



**If** test-condition **then** S<sub>1</sub>  
**else** S<sub>2</sub>  
**endif**

## Example 1.4.1

- Pseudo-code for the flowchart beside it:

**Input** A

**if**  $A > 0$  **then**

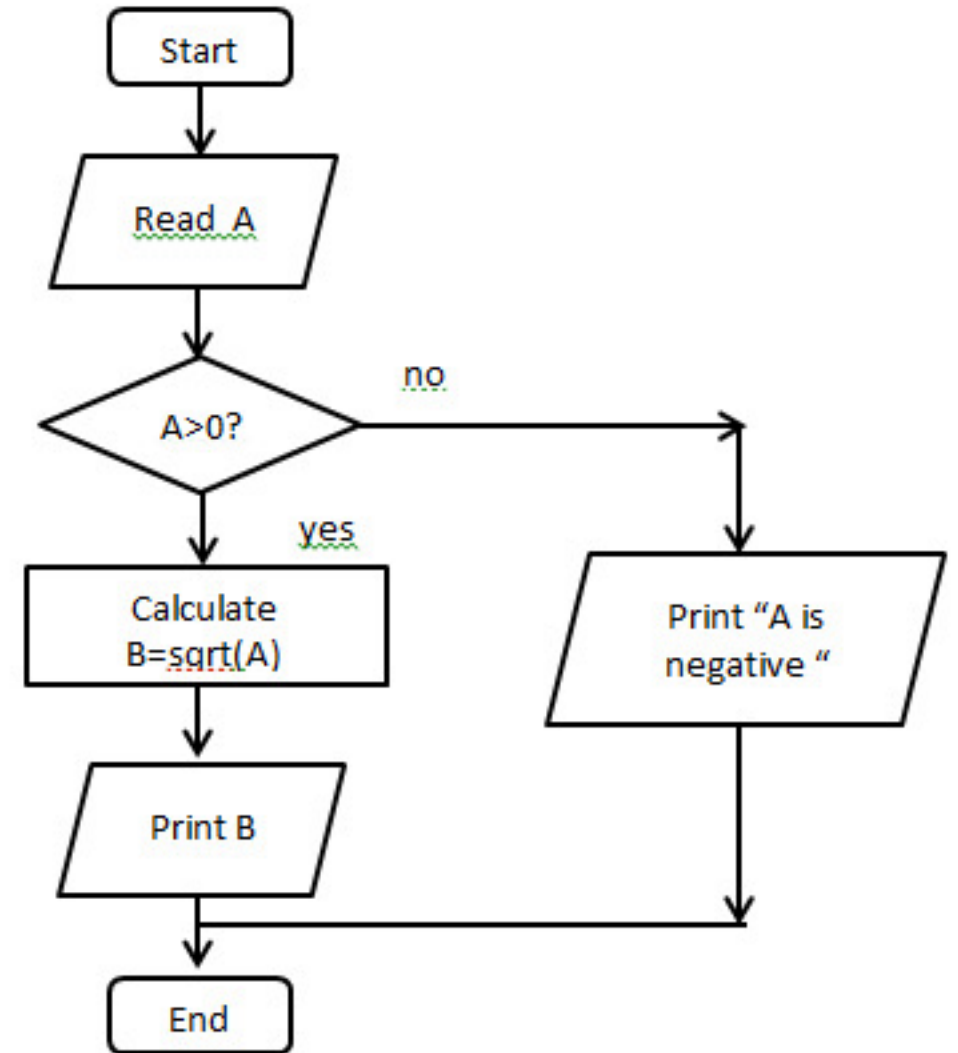
**calculate**  $B = \text{sqrt}(A)$

**print** B

**else**

**print** "A is negative"

**endif**



## Example 1.4.2

- Solving the quadratic equation  $ax^2 + bx + c = 0$
- Pseudo-code of the program:

**Input** a, b, c

**Calculate**  $del = b^2 + 4.a.c$

**if**  $del > 0$  **then**

$x1 = (-b + \text{sqrt}(del))/(2.a)$

$x2 = (-b - \text{sqrt}(del))/(2.a)$

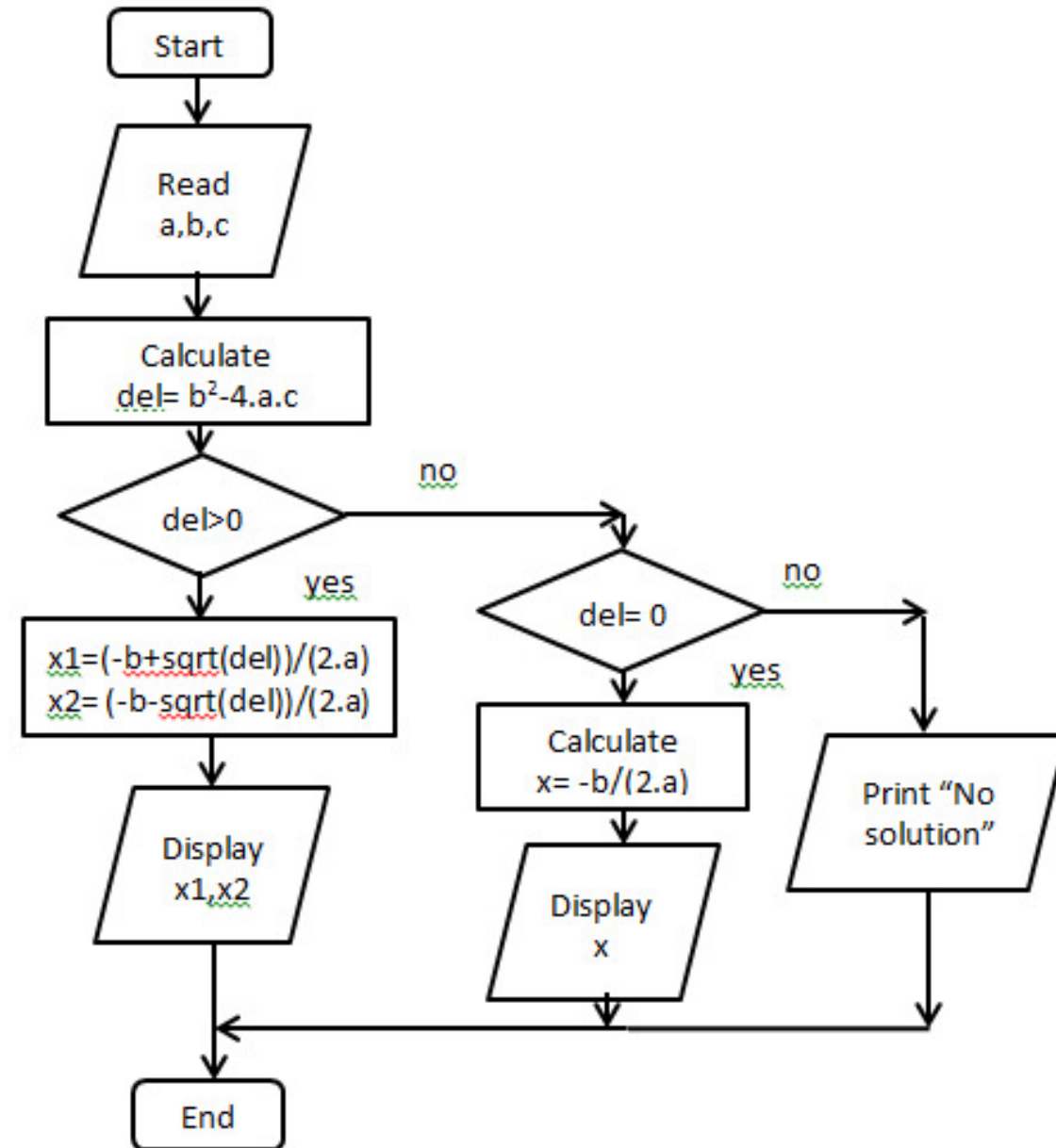
print x1, x2

**else if**  $del = 0$  **then**

$x = -b/(2.a)$

**else** print "No solution"

**endif**



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---

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# Loops on Algorithms

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- Many problems require repetition capability, in which the same calculation or sequence of instructions is *repeated*, over and over, using different sets of data.
- Loop is a very important concept in programming.



## Example 1.5.1

- Write a program to do the task: Print a list of the numbers from 4 to 9, next to each number, print the square of the number.
- Note:  $\text{NUM} \leftarrow \text{NUM} + 1$  means old value of  $\text{NUM} + 1$  becomes new value of  $\text{NUM}$ .
- The algorithm can be described in pseudocode as follows:

$\text{NUM} \leftarrow 4$

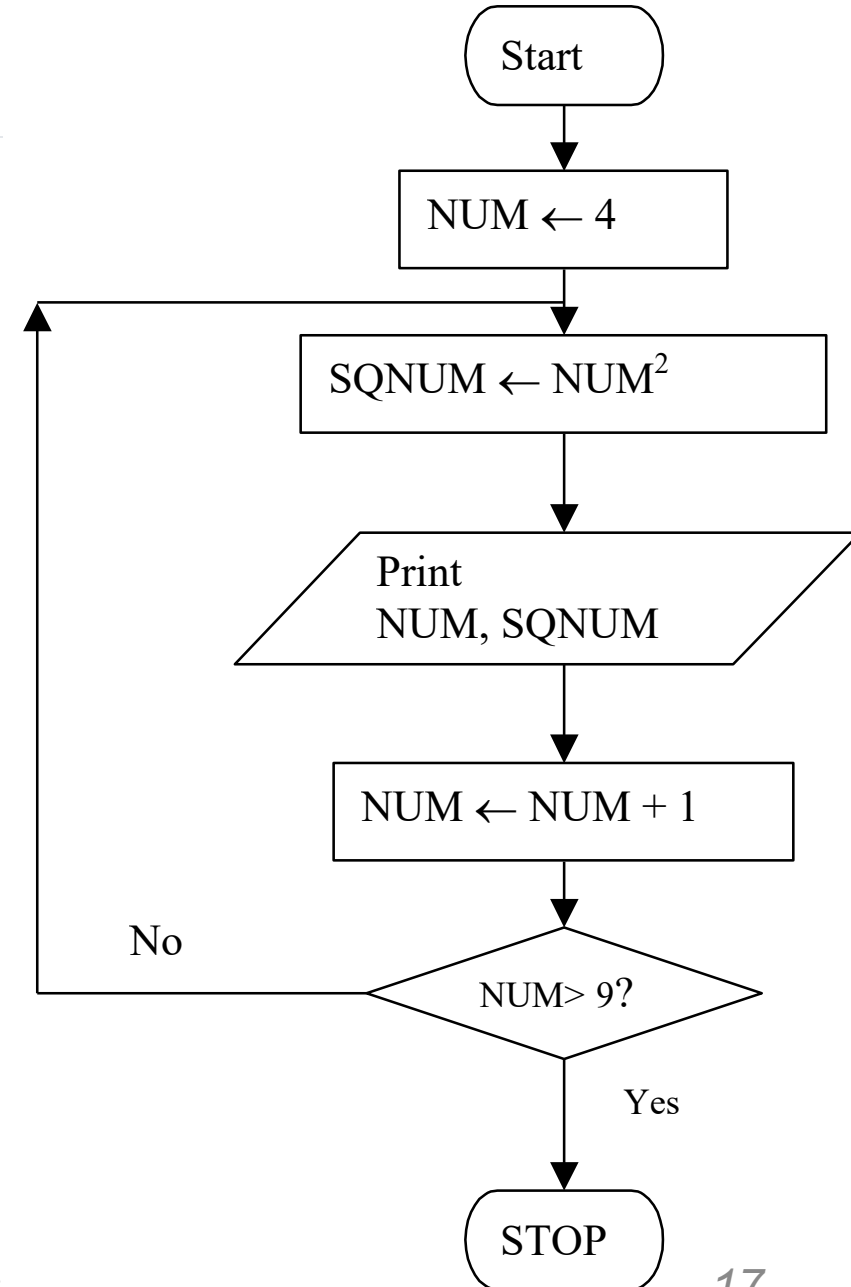
**do**

$\text{SQNUM} \leftarrow \text{NUM}^2$

Print  $\text{NUM}$ ,  $\text{SQNUM}$

$\text{NUM} \leftarrow \text{NUM} + 1$

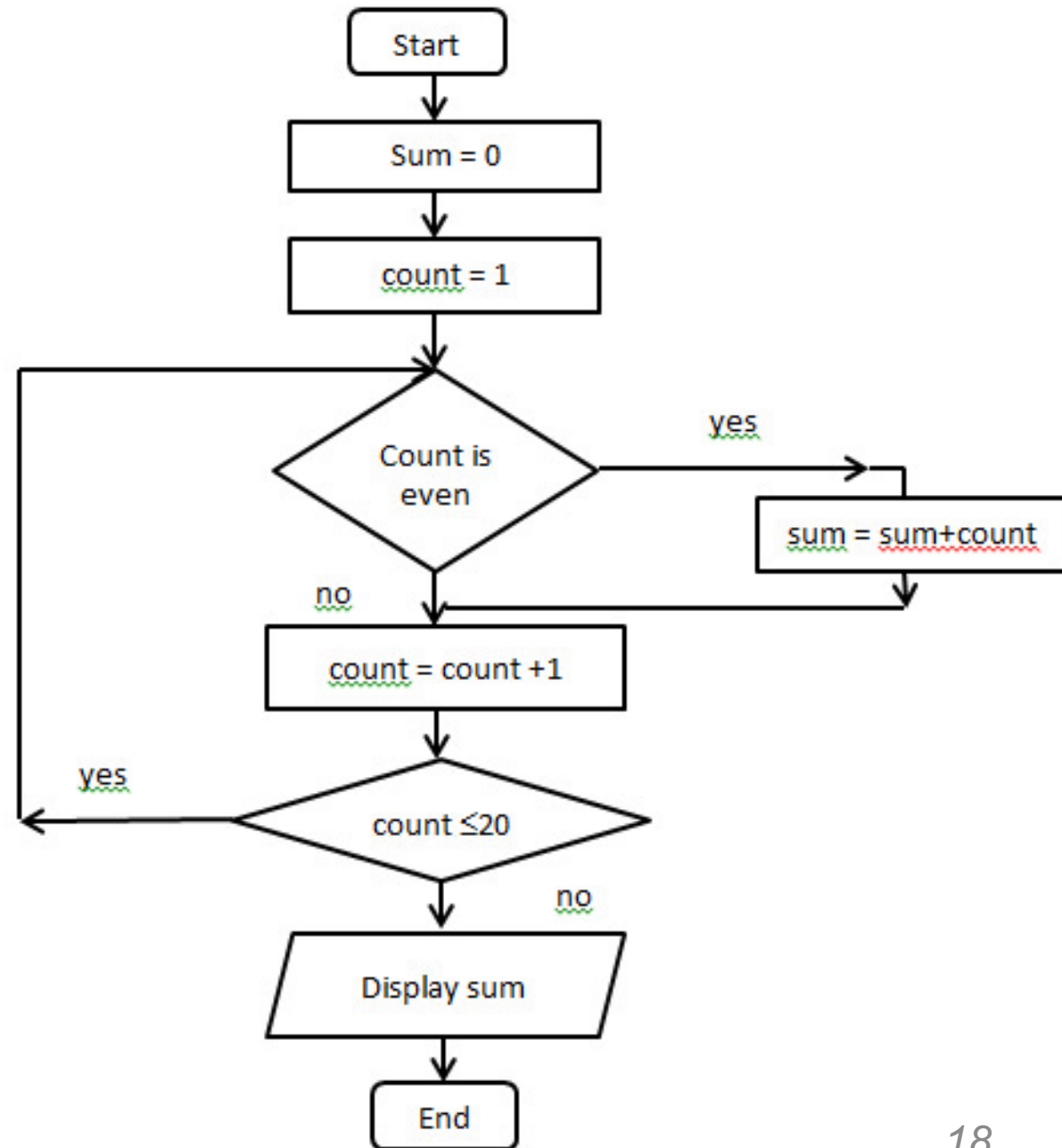
**while** ( $\text{NUM} \leq 9$ )



## Example 1.5.2

- Write a program to do the task: sum all the even numbers between 1 and 20 inclusive and then displays the sum.
- The algorithm can be described in pseudocode as follows:

```
sum = 0
count = 1
do
    if count is even then
        sum = sum + count
    endif
    count = count + 1
while count <= 20
Display sum
```



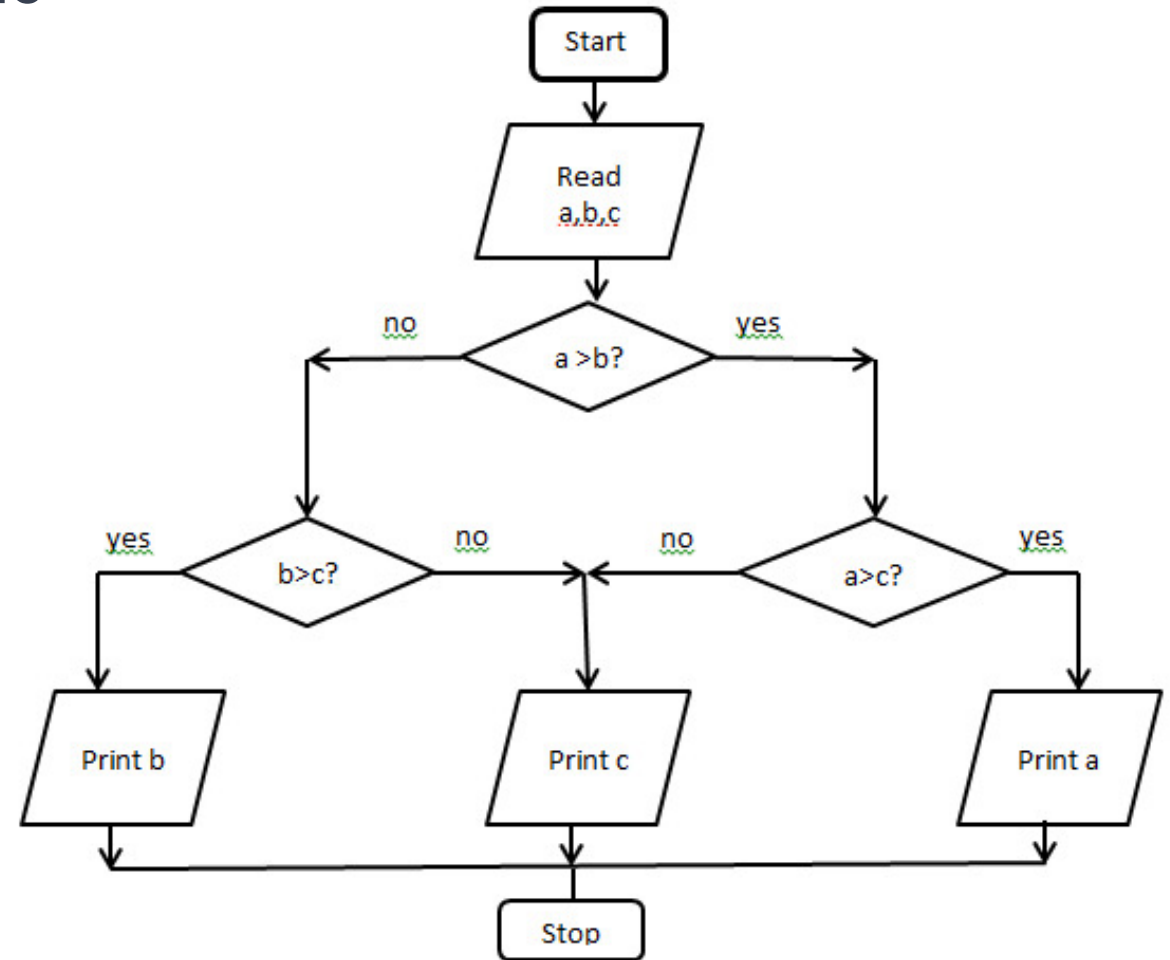
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# Exercise 1

- Given flowchart of program finding the largest among 3 different numbers.
- Write the pseudo-code.



# Exercise 1

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- Answer:

```
Input a,b,c
if a > b then
    if a > c then
        print a
    endif
else
    if b > c then
        print b
    else
        print c
    endif
endif
```

## Exercise 2

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- Given pseudo-code of program finding the sum of 5 numbers. Draw the flowchart.

1. Initialize  $\text{sum} = 0$  and  $\text{count} = 0$

2. Enter  $n$

3. Find  $\text{sum} + n$  and assign it to  $\text{sum}$

4. Increment  $\text{count}$  by 1

5. Is  $\text{count} < 5$ ?

6. If YES go to step 2

else

Print  $\text{sum}$

## Exercise 3

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- Write pseudo-code and draw a flowchart to read length and width of a rectangle and calculate its area and perimeter.



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# Thank you!